



EVERYDAY CHEMISTRY

⇒ CHEMISTRY IS ALL AROUND US! ⇐



Explore molecules, understand changes, improve life! ❤️

1

CHEMICAL CHANGES

A chemical change produces one or more new substances with different properties. It is usually permanent and not easily reversible.

EXAMPLES



Cooking an egg
(Irreversible)



Burning of candle
(Irreversible)



Rusting of iron
(Irreversible)

CHEMICAL REACTION

A chemical reaction is a process in which one or more substances (reactants) change into new substance(s) (products) by rearrangement of atoms.

REACTANTS



+



PRODUCTS



Example: Vinegar + Baking soda → Carbon dioxide + Water + Sodium acetate

PHYSICAL CHANGES

A physical change changes only the physical properties (size, shape, state) of a substance, not its chemical composition. It is usually reversible.

EXAMPLES



Melting of ice



Cutting of paper



Evaporation of water

2

ACIDS, BASES AND SALTS

ACIDS

Substances that taste sour, turn blue litmus red, and release hydrogen ions (H^+) in water.

EXAMPLES



Lemon + Vinegar

BASES

Substances that taste bitter, feel slippery, turn red litmus blue, and release hydroxide ions (OH^-) in water.

EXAMPLES



Soap + Baking Soda Solution

SALTS

Substances formed by the reaction of an acid and a base. They are usually crystalline and soluble in water.

EXAMPLE



Common Salt
(Sodium Chloride)

3

INDICATORS: NATURAL AND SYNTHETIC

NATURAL INDICATORS

INDICATOR (SOURCE)	NEUTRAL (pH 7)	ACIDIC (pH < 7)	BASIC (pH > 7)	USED IN
Turmeric (Curcuma longa)	Yellow	Yellow (no change)	Reddish Brown	Kitchen (foods, home remedies)
Red Cabbage (Anthocyanin)	Purple	Pink / Red	Green / Blue	Food, Education Labs
Hibiscus Flower	Pink	Dark Pink / Red	Green	Drinks, Education

SYNTHETIC INDICATORS

INDICATOR	NEUTRAL (pH 7)	ACIDIC (pH < 7)	BASIC (pH > 7)	USED IN
Litmus	Purple	Red	Blue	Labs, Schools, Industries
Phenolphthalein	Colorless	Colorless	Pink	Labs (Titrations)
Methyl Orange	Orange	Red	Yellow	Labs, Industries

4

PH PAPER DIAGRAM

20% →

← 20%



★ Remember: Acids donate H^+ ions, Bases donate OH^- ions, and neutral solutions have equal H^+ and OH^- ions. ★

Higher pH
= More Basic

